Name: _____ Mark: /30

Practical Assessment: Using Precipitation Reactions to Identify Solutions

This task is designed to test your practical skills as well as your skills of deduction.

Your Task

Your task is to identify four unknown solutions. You will do this by reacting each solution with three known solutions, recording your observations of the reactions and using your solubility table to identify the four unknowns. The unknown solutions, in no particular order are copper sulphate, sodium chloride, sodium hydroxide and potassium iodide.

Equipment

- Small test tubes x6 (you will need to rinse them between tests)
- Solutions of Silver nitrate, Lead nitrate and Calcium nitrate
- Unknown solutions A-D
- Test tube rack
- Safety glasses
- Lab coat

Method

1. Put about 10 drops of unknown A into 3 test tubes.
2. To the first test tube add 5 drops of silver nitrate, to the second test tube add 5 drops of lead nitrate and to the third add 5 drops of calcium nitrate.
3. Record your observations in the table.
4. Repeat steps 1, 2 and 3 for each unknown.

Results

/12

Unknown	Silver nitrate	Lead nitrate	Calcium nitrate
A			
B			

C			
D			

Processing of Results

Use the table of solubility rules to work out what each unknown solution is. /4

copper sulphate sodium chloride sodium hydroxide potassium iodide

Solution A: _____ Solution B: _____

Solution C: _____ Solution D: _____

There are 2 hints available to help you figure out the unknowns, however you will lose 1 mark for each hint you use.

(Hints used 0 1 2)

Questions

Use the solubility rules to predict the precipitate formed when these pairs of solutions are mixed. If no solid is formed, write no reaction. /8

1. Silver nitrate and sodium chloride

2. Copper sulphate and sodium nitrate

3. Mercury (I) nitrate and potassium iodide

4. Calcium nitrate and lithium carbonate

5. Barium nitrate and sodium sulphate

6. Copper sulphate and sodium hydroxide

7. Sodium hydroxide and lead nitrate

8. Barium nitrate and lead nitrate

Working in the Laboratory

Appropriate use of safety equipment

/2

Appropriate behaviour

/2

Workstation cleaned properly

/2